MILITARY SPECIFICATION
SURFACE ACOUSTIC WAVE DEVICES,
GENERAL SPECIFICATION FOR

This specification is approved for use by USA Electronic Research and Development Command,Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the general requirements for surface acoustic wave (SAW) devices with or without passive impedance matching components packaged in a hermetic enclosure. The specification establishes the quality and reliability assurance requirements which must be met in the acquisition of SAW devices. Detailed requirements, specific characteristics and other provisions which are sensitive to the particular device application shall be specified in the applicable device specification.

1.2 Classification

1.2.1 Military part number. SAW devices specified herein (see 3.1) shall be identified by a military part number which shall consist of the basic number of the specification sheet and a coded number. The part number shall be coded to provide information concerning the intended use of the device and product assurance level. The military part number shall be in the following format:

M49433/001	BP	B
Military designator (1.2.1.1)	Intended Use (1.2.1.2)	Product Assurance Level (1.2.1.3)

1.2.1.1 Military designator. The military designator consists of the specification number and the applicable detail specification.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Electronics Research and Development Command, ATTN: DELET-R-S, Fort Monmouth, NJ 07703 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5999

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

1.2.1.2 Intended use. The intended use is identified by two letters in accordance with table I.

Table I. Intended Use

Symbol	Intended Use
BP	Bandpass Filter
RE	Resonator
CR	Correlator
CV	Convolver
DL	Delay Line
MU	Multiplexer (filter bank)
DF	Dispersive Filter

1.2.1.3 Product assurance level. This specification provides for 2 levels of quality and reliability assurance, class B, and class S. The product assurance level is identified by a single letter in accordance with table II.

Table II. Product assurance levels

Symbol	Level
B	Class B
S	Class S (space applications)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.1) the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specification and Standards DODISS specified in the solicitations, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-T-90 - Tape, Pressure Sensitive Adhesive (Cellophane and Cellulose Acetate).

MILITARY

MIL-Q-9858 - Quality Program Requirements.

MIL-M-55565 - Microcircuits, Packaging of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

MIL-STD-1285 - Marking for Electrical and Electronic Parts.

MIL-STD-45662 - Calibration Systems Requirements.

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable detail specification. In the event of any conflict between the requirements of this specification and the detail specification, the latter shall govern.

3.2 Qualification. SAW devices furnished under this specification shall be the products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.5 and 6.2).

3.2.1 Traceability documentation. Manufacturers and any other suppliers who offer the product described by this specification shall provide written certification signed by a responsible official, (1) that the product being supplied is the product listed on the Qualified Product List and (2) of the source (including plant location and mailing address) of the manufacturer of the product. This traceability documentation or copy thereof must be certified and forwarded by each successive supplier handling the product.

3.2.2 Change of qualified product. Changes in design, materials, or processes for any SAW devices must be processed in accordance with established change control procedures for all affected documents. The manufacturer shall notify the qualifying activity prior to the implementation of any major change of the product or product assurance program which may affect performance, quality, reliability, or interchangeability. Such notification shall include a description of the proposed change and a suggested test plan or test data to demonstrate that the change will not adversely affect performance, quality, reliability or interchangeability and that the changed product will continue to meet the specification requirements. The qualifying activity will review test plan or test data and advise the manufacturer of the testing required to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

obtain approval of the proposed change. Upon completion of the prescribed tests, the results shall be submitted to the qualifying activity for review. Based upon the review, the qualifying activity will approve or disapprove the change and notify the manufacturer. At the manufacturer's option, devices incorporating the change may be manufactured and tested prior to approval; however, all shipment of these changed devices shall be withheld until approval is granted by the qualifying activity. Changes representative of those which are subject to this requirement are any basic design changes in the manufacturing processing such as:

a. Mask changes or redesign which alter die size or circuit dimensions, spacing, or isolation.

b. Metallization change (material), deposition technique.

c. Change in die or substrate attach material.

d. Wire bond method changes.

e. Changes in wire material composition and dimensions.

f. Package or lid structure and material changes.

g. Seal technique changes (materials or sealing process).

h. Implementation procedures for internal visual inspection and other test methods.

i. Changes to process flow charts, assembly flow chart, and testing flow charts.

j. Substrate material.

k. Design change of die or other elements.

l. Device trimming methods and processes other than the SAW device.

m. Wafer fabrication move from one line or building to another.

n. Assembly operation move from one line or building to another.

o. Test facility (with laboratory suitability) move from one facility or building to another.

p. Qualification or quality conformance inspection procedures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

3.3 Screening. All SAW devices to be delivered or submitted for qualification or submitted for quality conformance in accordance with this specification shall have been subjected to, and passed, all the screening tests detailed in method 5008 of MIL-STD-883 except that burn-in shall not be done, for the type of SAW device and product assurance level specified. Sampling inspections shall not be an acceptable substitute for any specified screening test.

3.4 Materials. The materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the SAW devices to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be classified as a guaranty of the acceptance of the finished product.

3.4.1 Metals. External metal surfaces shall be corrosion-resistant or shall be plated or treated to resist corrosion.

3.4.2 Other materials. External parts, or coatings including markings, shall be inherently non-nutrient to fungus and shall not blister, crack, outgas, soften, flow or exhibit defects that adversely affect storage, operation, or environmental capabilities of SAW devices delivered to this specification under the specified test conditions.

3.5 Design and construction. SAW device design and construction shall be in accordance with all the requirements specified herein and in the detail specification (see 3.1).

3.5.1 Design documentation. Design, topography, and schematic circuit information for all SAW devices supplied under this specification shall be submitted to the qualifying activity and shall be available in-plant for review by the procuring activity and the qualifying activity upon request. This design documentation shall be sufficient to depict completely the physical and electrical construction of the SAW devices supplied under this specification, and shall be traceable to the specific part, drawing or type numbers to which it applies, and to the production lot(s) and inspection lot codes under which SAW devices are manufactured and tested so that revisions can be identified.

3.5.1.1 Schematic diagrams. For SAW devices supplied under this specification, the actual schematic diagram(s) shall be provided sufficient to represent all electrical elements functionally designed into the SAW device together with their values (when applicable). For simple devices this shall be a complete detailed schematic circuit showing all functional elements and values.

3.5.2 Package. All devices shall be hermetically sealed in metal or ceramic (or combinations thereof) packages. In addition, packages for class S SAW devices shall have a metal body with hard glass seals or a ceramic body. The lids shall be welded, brazed, preform soldered, or glass frit sealed. Packages using glass frit seal shall have glass on the mating surfaces only, and the inside surfaces of the cavity shall not be coated with the seal glass.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

3.5.3 Metallization. The metal film (e.g. aluminum, gold, or chromium) shall be deposited on the polished piezoelectric substrates.

3.5.3.1 Adhesion of metal film. The adhesion of the metallic film to the crystal substrate shall remain intact without peeling, lifting or blistering (see 4.7.2).

3.5.4 Wafers. Circuits on the undiced wafer shall be visually checked (i.e. compared) for obvious defects (see 4.7.1.2).

3.5.5 Internal visual. The SAW device chip shall be examined after placement in the uncovered package for broken wire bonds, foreign material, scratches and other circuit imperfections (see 4.7.1.3).

3.5.6 Wafer dicing. When required to separate the multiple identical SAW device circuit chips fabricated on the same wafer, a precision dicing device which minimizes kerf loss, shall be used.

3.5.7 Internal wire bonding. Internal wire bonding shall be performed using metallurgical bonding techniques to effect reliable electrical connections (see 4.7.3). There shall be no evidence of loosening or breakage of wire bond connection.

3.5.8 External lead integrity. When tested as specified in 4.7.4 there shall be no evidence of breakage or loosening between the lead and the device body.

3.5.9 Device packaging. Packaging and sealing techniques shall be used so that the resultant hermetically sealed device shall be capable of meeting all environmental requirements specified. Unless otherwise specified (see 3.1), only properly cured expoxy, or low outgassing adhesive materials shall be used internally to the sealed device package.

3.6 Hermetic seal. When tested as specified in 4.7.5, SAW devices shall show no evidence of leakage.

3.7 Thermal shock. Each SAW device shall exhibit no evidence of electrical, mechanical or physical damage after being exposed to specified temperature cycling (see 3.1 and 4.7.6).

3.8 High temperature storage. When tested as specified in 4.7.7, SAW devices shall show no evidence of electrical, physical or mechanical damage.

3.9 Shock (specified pulse). When tested as specified in 4.7.8, SAW devices shall show no evidence of electrical, physical or mechanical damage.

3.10 Vibration (low frequency). When tested as specified in 4.7.9, SAW shall show no evidence of electrical, mechanical or physical damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

3.11 Moisture resistance. When tested as specified in 4.7.10, SAW devices shall show no evidence of mechanical or physical damage (corrosion) and shall meet the electrical characteristics as specified in 3.16 and the applicable detail specification (see 3.1).

3.12 Life. When tested as specified in 4.7.11, SAW devices shall show no evidence of mechanical or physical damage and shall meet the electrical characteristics as specified in 3.16 and the applicable detail specification (see 3.1).

3.13 Resistance to solvents. When SAW devices are tested as specified in 4.7.12, there shall be no evidence of physical damage to the package and the marking shall remain legible.

3.14 Solderability. When applicable (see 3.1), leads shall be solderable (see 4.7.13).

3.15 Electrostatic discharge. When SAW devices are found to be sensitive by the electrostatic discharge (ESD) sensitivity test (see 4.7.14), the devices represented by the test shall be considered category A devices.

3.16 Electrical requirements (4.7.15). When tested as specified in 4.7.15, SAW devices shall meet the electrical requirements of the detail specification (see 3.1) and shall include, as applicable, the parameters listed below:

a. Center frequency of operation.

b. Bandwidth.

c. Insertion loss.

d. Feedthrough suppression.

e. Spurious echo suppression.

f. Voltage standing wave ratio (VSWR).

3.17 Marking.

3.17.1 JAN and J marking. The United States Government has adopted, and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of military specifications. Accordingly, items procured to, and meeting all of the criteria specified herein and in applicable detail specifications shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

line above or below the part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable detail specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets or detail specifications, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products procured to contractor drawings or specifications. The United States Government has obtained Certificate of Registration No. 504,860 for the certification mark "JAN".

3.17.2 Full marking. Each SAW device shall be marked in accordance with Method 1 of MIL-STD-1285 and the applicable detail specification. If any special marking is required (see 6.1), it shall in no way interfere with the marking required herein and shall be visibly separate therefrom. The following marking shall be placed on each SAW device unless otherwise specified:

a. Military part number (see 1.2.1).

b. Date code or lot symbol.

c. Manufacturer's source code.

d. JAN certification.

e. Serialization, when applicable (see 3.17.5).

f. ESD in accordance with MIL-STD-1285, when applicable (see 3.15).

3.17.3 Minimum marking. When the physical size of the SAW device precludes the full marking specified in 3.17.2, the minimum marking required shall be as specified on the applicable detail specification. In those cases where full marking requirements are not on the SAW device, the full marking shall be marked on the unit package (see 5.1.2).

3.17.4 Marking location and sequence. Unless otherwise specified, the "JAN" or "J" certification mark, the part number, inspection lot identification code, and serialization (where applicable), shall be located on the top surface of flat packages or dual-in-line configurations and on either the top or the side of cylindrical packages. The balance of the markings may be placed in any suitable location so as to perform their required functions and not interfere with the other markings except that the ESD identifier shall be marked on the top.

3.17.5 Serialization. Prior to the first recorded electrical measurement in screening each class S SAW device, and, when specified each class B SAW device, shall be identified with a unique serial number assigned consecutively within the inspection lot. This serial number allows traceability of test results down to the level of the individual SAW device within that inspection lot (see 4.5.1). For class S, inspection lot records shall be maintained to provide traceability from the serial number to the specific wafer lot from which the devices originated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

3.18 Workmanship. SAW devices shall be manufactured, processed, and tested in a careful and workmanlike manner in accordance with good engineering practice, and with the production practices, workmanship instructions, inspection and test procedures, prepared by the manufacturer in fulfillment of the product assurance program.

3.18.1 Rework provisions. All rework permitted on SAW devices procured under this specification shall be accomplished in accordance with procedures and safeguards documented and available for review by the qualifying or procuring activity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Reliability assurance program (class S only). A reliability assurance program shall be established and maintained in accordance with MIL-Q-9858. The following shall apply:

a. The manufacturer shall be able to identify the time period during which the final production operation was performed on each item of product prior to final test. The date or lot code marked on each part shall be identified to an inspection lot.

b. The requirement for traceability of materials as defined in 4.3 shall apply and be documented in the baseline.

4.1.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection.

a. Materials inspection (see 4.3).

b. Qualification inspection (see 4.5).

c. Quality conformance inspection (4.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials used in fabricating the SAW devices are in accordance with the applicable specification and requirements prior to fabrication.

4.4 Inspection conditions. Unless otherwise specified herein, or in the detailed specification, all inspections shall be made in accordance with the general requirements of MIL-STD-883.

4.5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.2) on sample units produced with equipment and procedures normally used in production.

4.5.1 Sample. The number of sample SAW devices comprising a sample of SAW devices to be submitted for qualification inspection shall be as specified in table III or as specified in the detail specification (see 3.1). All qualification test samples shall be serialized prior to qualification testing for class S.

4.5.2 Test routine. Qualification samples shall be subjected to the tests of table III. All sample units shall be subjected to tests of group I. The sample SAW devices shall then be divided into 5 groups and subjected to the tests of groups II-VI in the order shown for the specific class of SAW device being qualified (class B or S).

4.5.2.1 ESD test routine. An additional 3 samples shall be subjected to group VII testing. Group VII shall be performed only at initial qualification unless there is a change in design, materials, or process (see 3.15).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

Table III. Qualification inspection

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH	CLASS B		CLASS S	
			No. to be Tested	No. of Failures	No. to be Tested	No of Failures
Group I						
Design and construction	3.5.2-3.5.6 3.5.9-3.5.10	4.7.1, 4.7.2	ALL UNITS (20)	2	ALL UNITS (30)	0
Marking	3.17					
Internal wire bonding	3.5.7	4.7.3				
External lead integrity	3.5.8	4.7.4				
Electrical characteristics	3.16	4.7.15				
Group II						
Solderability (when applicable)	3.14	4.7.13	4	1	6	
Group III						
High temperature storage	3.8	4.7.7	4	1	6	
Electrical characteristics	3.16	4.7.15				
Group IV						
Life	3.12	4.7.11	4	1	6	1
Electrical characteristics (Final)	3.16	4.7.15				
Group V						
Hermetic seal	3.6	4.7.5	4	1	6	
Group VI						
Vibration	3.10	4.7.9	4	1	6	
Shock (specified pulse)	3.9	4.7.8				
Thermal shock (10 cycles)	3.7	4.7.6				
Moisture resistance	3.11	4.7.10				
Electrical characteristics (Final)	3.16	4.7.15				
Group VII						
Electrostatic discharge	3.15	4.7.14	3 1/		3 1/	

1/ Units to be tested for ESD shall be in addition to the 20(30) units (see 4.5.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

4.5.3 Failures. Failures in excess of those allowed in table III (or as specified in the detail specification, see.3.1) shall be cause for refusal to grant qualification.

4.5.4 Retention of qualification.

4.5.4.1 Class B devices. To retain qualification, the contractor shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for group A inspection indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.

b. A summary of the results of tests performed for periodic inspection (group B), including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 12-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

If the summary of test results of group A and group B inspections indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 12-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit his qualified products to testing in accordance with the qualification inspection requirements.

4.5.4.2 Class S devices. To retain qualification for class S Devices, the manufacturer shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall include the following:

a. A list of all device types for which the manufacturer has qualification, showing the total number of lots completing quality conformance inspection during the reporting period.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

b. Certification that the manufacturer still has the capabilities and facilities necessary to produce these items to the qualified tests.

c. The number of lots which failed lot-by-lot testing.

d. For failed lots the following information is required:

(1) Identification of the lots by inspection lot identification, date and lot code, part number, and date of completion of group B testing.

(2) The results of groups A and B testing (i.e., pass or fail).

(3) For failed groups, the number and mode of all failures.

(4) All lots withdrawn shall be counted as failed lots in the report.

If the summary of test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 12-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit his qualified products to testing in accordance with the qualification inspection requirements.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A, B and C inspections.

4.6.2 Inspection lot class B devices. An inspection lot shall be as defined in MIL-STD-105 and shall consist of the devices of a single part number, produced under essentially the same conditions and offered for inspection at one time.

4.6.2.1 Inspection lot class S devices. An inspection lot shall be as defined in MIL-STD-105 and shall be of one design as specified on the applicable detail specification sheet. As a minimum, the devices shall be processed as a single lot through all manufacturing steps on the same equipment and identified with the same date and lot code. The manufacturer shall maintain records on each lot date code.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

4.6.3 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table IV. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table IV. Major and minor defects shall be as defined in MIL-STD-105.

Table IV. Group A Inspection

EXAMINATION OR TEST	REQUIRED PARAGRAPH	TEST METHOD PARAGRAPH	AQL (Percent Defective)			
			CLASS B		CLASS S	
			MAJOR	MINOR	MAJOR	MINOR
Subgroup 1 - WAFERS						
Adhesion of metallic film	3.5.3.1	4.7.2	1.0%	2.5%	1.0%	2.5%
Subgroup 2 - SAWD'S						
Design and construction	3.5.2-3.5.6 3.5.9-3.5.10	4.7.1, 4.7.2				
Marking	3.17	---	1.0%	2.5%	0.25%	1.0%
Internal wire bonding	3.5.7	4.7.3				
External lead integrity	3.5.8	4.7.4				
Hermetic seal	3.6	4.7.5				

4.6.3.1 Rejected lots class B. If an inspection lot is rejected, the manufacturer may rework the lot to correct the defects, or screen out the defective units and resubmit the reworked lot for inspection. Resubmitted lots shall be inspected using tightened inspection of MIL-STD-105.

4.6.3.2 Rejected lots class S. Rejected inspection lots may be resubmitted for Government acceptance only if the manufacturer performs 100 percent inspection on SAW devices of the lot for those characteristics which were defective and resulted in rejection of the lot and removes all defective units and resubmits the lot for inspection. Resubmitted lots shall be kept separate from new lots, and shall be clearly identified as resubmitted lots. Resubmitted lots shall be inspected using the tightened inspection procedure of MIL-STD-105 and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed.

4.6.4 Group B inspection. Group B inspection shall consist of the inspections specified in table V. They shall be performed on sample units that have been subjected to and have passed the group A inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

Table V. Group B inspection

Examination or Test	Requirement Paragraph	Method Para	AQL (Percent Defective)	INSP LEVEL
Electrical characteristics Center frequency of operation Bandwidth Insertion loss Feedthrough suppression Spurious echo suppression Voltage standing wave ratio (VSWR) (when applicable)	3.16	4.7.15	2.5%	S-4

4.6.4.1 Rejected lots (class B). If an inspection lot is rejected, the contractor may withdraw the particular lot once, screen out defectives, and reinspect once. Such lots shall be kept separate from new lots and shall be clearly identified as reinspected lots. Rejected lots shall be reinspected using tightened inspection.

4.6.4.1.1 Disposition of samples class B. Sample units which have passed all the group B inspections may be delivered on the contract or purchase order if the lot is accepted and the sample units are still within the electrical tolerances.

4.6.4.2 Rejected lots (class S). If an inspection lot fails to pass group B inspection, the inspection lot shall not be delivered on the contract or purchase order. The manufacturer shall notify the qualifying activity and the cognizant inspection activity of the failure and take corrective action on the materials or processes or both as warrented. Rejected lots shall be reinspected using tightened inspection.

4.6.5 Group C inspection. Group C inspection shall be performed once each 6 months and shall consist of the tests specified in table VI, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed groups A and B inspections, except that samples for the solderability test may be selected from samples that have failed group A and group B testing.

4.6.5.1 Sampling plan. 20 sample units shall be selected from each 6 months production of the same function and class for each Group C test. The sample units shall be subjected to the tests shown in table VI. Not more than one failure shall be allowed for a single group of 20 samples.

4.6.5.2 Disposition of samples. When specified by the procuring activity, SAW devices subjected to group C inspection shall be delivered on the contract or order (see 6.1), except that samples subjected to the solderability test shall not be delivered.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

4.6.5.3 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

Table VI. Group C inspection.

Examination or test	Requirement Paragraph	Test paragraph
Subgroup 1		
High temperature storage	3.8	4.7.7
Hermetic seal	3.6	4.7.5
Subgroup 2		
Life	3.12	4.7.11
Electrical characteristics (final)	3.16	4.7.15
Subgroup 3		
Vibration	3.10	4.7.9
Shock (specified pulse)	3.9	4.7.8
Thermal shock (10 cycles)	3.7	4.7.6
Moisture resistance	3.11	4.7.10
Electrical characteristics (final)	3.16	4.7.15
Subgroup 4		
Solderability (when applicable)	3.14	4.7.13

4.6.6 Inspection of packaging. SAW device packaging shall be inspected in accordance with the applicable paragraphs of MIL-M-55565. The requirements for radiation protection are not applicable.

4.7 Methods of inspection.

4.7.1 Visual inspection.

4.7.1.1 External visual and mechanical inspection. SAW devices shall be examined to verify that the materials, design, construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.5, 3.17 and 3.18).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

4.7.1.2 Wafer (see 3.5.4). An enlarged 5X to 30X graphic or photographic representation shall be used for visual comparison to ensure sameness of device patterns. The viewing magnification shall be sufficient to provide the required resolution to detect mask number and gross defects.

4.7.1.3 Internal visual (3.5.5). An examination for circuit imperfections (i.e. broken wire bonds, dirt, scratches, etc.) shall be performed under 20X magnification.

4.7.2 Adhesion of metal film (3.5.3.1). A one-inch strip of pressure sensitive cellophane tape, conforming to type I, class A of L-T-90, shall be applied to the metallized surface, adhesive side down, using firm hand pressure. The tape shall be removed with one abrupt motion; the adhesive side shall then be examined for detached particles of metallic film. The test shall be performed on a sample substrate from the same metallization lot as the wafers under inspection. If the sample substrate fails, the entire metallization lot is deemed to have failed.

4.7.3 Internal wire bonding (3.5.7). For class S devices,internal wire bonding shall be tested in accordance with method 2023 of MIL-STD-883.

4.7.4 External lead integrity (3.5.8). Devices shall be tested in accordance with method 2004 of MIL-STD-883. Test condition A shall be used.

4.7.5 Hermetic seal (3.6). Devices shall be tested in accordance with method 1014 of MIL-STD-883. Test condition A1 or A2, or B,as applicable, shall be used. For gross leak, test condition C shall be used.

4.7.6 Temperature cycling (3.7). Devices shall be tested in accordance with method 1010 of MIL-STD-883, test condition A (10 cycles).

4.7.7 High temperature storage (3.8). Devices shall be tested in accordance with method 1008 of MIL-STD-883; the following details shall apply:

a. Test condition - B (+125°C unless otherwise specified [see 3.1])

b. Test duration - 72 hours

c. At the end of exposure time, the devices shall be allowed to stabilize at room temperature and shall meet the requirements of 3.16.

4.7.8 Shock (specified pulse) (3.9). Devices shall be tested in accordance with method 213 of MIL-STD-202; the following details shall apply:

a. Test condition - A

b. Method of mounting - devices shall be rigidly mounted.

4.7.9 Vibration (low frequency (3.10). Devices shall be tested in accordance with method 201 of MIL-STD-202; the devices shall be rigidly mounted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

4.7.10 Moisture resistance (3.11). Devices shall be tested in accordance with method 1004 of MIL-STD-883. Before final measurements are made, all units shall be removed from the test chamber and stabilized at room temperature. The devices shall be visually inspected for corrosion and marking. The devices shall meet the electrical requirements as specified in 3.16 and the applicable detail specification sheet. Degradation limits shall be as specified in the detail specification (see 3.1).

4.7.11 Life (at elevated ambient temperature) (3.12). Devices shall be tested in accordance with method 108 in MIL-STD-202. The following details and exceptions shall apply:

a. Test temperatures (unless otherwise specified [see 3.1]) - 100°C + 10°C
- 5°C

b. Test condition (unless otherwise specified [see 3.1]) - D

c. Final measurements - Before measurements are made, all units shall be removed from the test chamber and stabilized at room temperature. The devices shall meet the electrical requirements as specified in 3.16 and the applicable detail specification sheet. Degradation limits shall be specified in the detail specification (see 3.1).

4.7.12 Resistance to solvents (3.13). Devices shall be tested in accordance with method 215 in MIL-STD-202. The spray method for the fourth group shall be used.

4.7.13 Solderability (3.14). Devices shall be tested in accordance with method 2003 of MIL-STD-883. The following shall be specified in the detail specification,as applicable (see 3.1):

a. The number of terminations of each part to be tested (see 3.1).

b. Special preparation of the terminations, if applicable (see 3.1).

c. Depth of immersion if other than specified (see 3.1).

d. Solder dip.

e. Examination of terminations.

f. Measurements after test, where applicable.

g. Solder composition, flux, and temperature, if other than those specified.

h. Number of cycles, if other than one. Where more than one cycle is specified to test the resistance of the device to heat as encountered in multiple solderings, the examinations and measurements required shall be made at the end of the first cycle and again at the end of the total number of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

cycles applied. Failure of the device on any examination and measurement at either the one-cycle or the end point shall constitute failure to meet this requirement.

4.7.14 Electrostatic discharge (3.15). Category B devices shall be tested in accordance with method 3015 of MIL-STD-883 and the applicable detail specification (see 3.1).

4.7.15 Device electrical characteristics (3.16). Devices shall be tested in accordance with established test methods or as specified on the applicable detail specification sheet. End point limits shall be as specified (see 3.1).

5. PACKAGING.

5.1 Packaging requirements. SAW devices shall be packaged in accordance with the applicable requirements of MIL-M-55565. In addition, devices which have been determined to require electrostatic discharge protection (category A), shall be packaged in conductive material or packaged in antistatic material with an external conductive field shielding barrier. Packaging other than a., b., or c. below shall require approval of the qualifying activity:

a. Unit container consisting of 3 layers with an antistatic inner surface and one of the following options: electrostatic field shielding exterior layer and an insulative intermediate layer or an insulative exterior layer and a conductive electrostatic field shielding intermediate layer.

b. Conductive noncorrosive rail with noncorrosive and conductive or antistatic foam plugs at both ends of each rail to prevent movement.

c. Antistatic noncorrosive rail with noncorrosive and conductive or antistatic foam plugs at both ends of each rail to prevent movement. Antistatic rails shall be packaged in conductive, electrostatic field shielding material.

5.1.2 Unit package marking. When applicable, the unit package marking specified in 3.17.3 shall be applied in accordance with the special marking provisions of MIL-STD-129.

6. NOTES.

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Title, number and date of the applicable detail specification sheet and the complete military part number (see 3.1).

c. Levels of preservation and packing required (see 5.1).

d. Special marking.

e. Delivery of devices subjected to group C inspection (see 4.6.5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

6.2 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is USA Electronics Research and Development Command, Ft. Monmouth, NJ. However, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), Dayton, Ohio 45444.

6.3 Terms and definitions.

a. Surface acoustic wave (SAW) device. A class of signal processing devices based on the use of elastic (Rayleigh) wave propagation on the surface of a piezoelectric solid so that a traveling electric field extends above the surface and can interact with metal electrodes on it.

b. Substrate. The supporting material upon which the SAW device is constructed.

c. Surface acoustic wave (SAW) transducer. A comb structure applied to the surface of a substrate consisting of interleaved metal electrodes whose function is to transform electrical energy into acoustic energy or vice versa by means of the piezoelectric effect.

d. SAW device functional type. SAW devices which are designed to perform the same type of basic circuit function (e.g. delay lines, bandpass filters, resonators, dispersive delay lines, multiplexers, correlators, and convolvers) and which are fabricated by use of the same basic construction and metallization.

e. Device type. The term device type refers to a single specific substrate/package configuration. Samples of the same device type will be electrically and functionally interchangeable with each other. The electrical and environmental limits will be the same for a given device type.

f. Final seal. That manufacturing operation which completes the enclosure of a device so that further internal processing cannot be performed without disassembling the device.

6.4 General Information

6.4.1 Center frequency -f_o. While this is a frequency domain parameter, for tap delay lines (TDL) and pulse compression (PC) devices (that are delivered as matched conjugate pairs) the parameter is measured in the time domain because of greater accuracy, less operator dependence, and relative ease. The time domain measurement is accomplished by summing the output response from the device under test (DUT) with the output of the generator which supplies the original signal using a differential preamplifier in the oscilloscope.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

When the two signal input amplitudes are exactly the same, but 180^{0} out of phase, cancellation takes place and the precise frequency can be read from the signal generator or synthesizer. Figure 1 illustrates how this is accomplished for PC and TDL devices. When both signals are inserted into the scope preamplifier simultaneously and the center frequency of the signal source and the line stretcher are alternately adjusted for optimum null in the center of the presentation the exact center frequency can be read from the center frequency readout of the synthesizer.

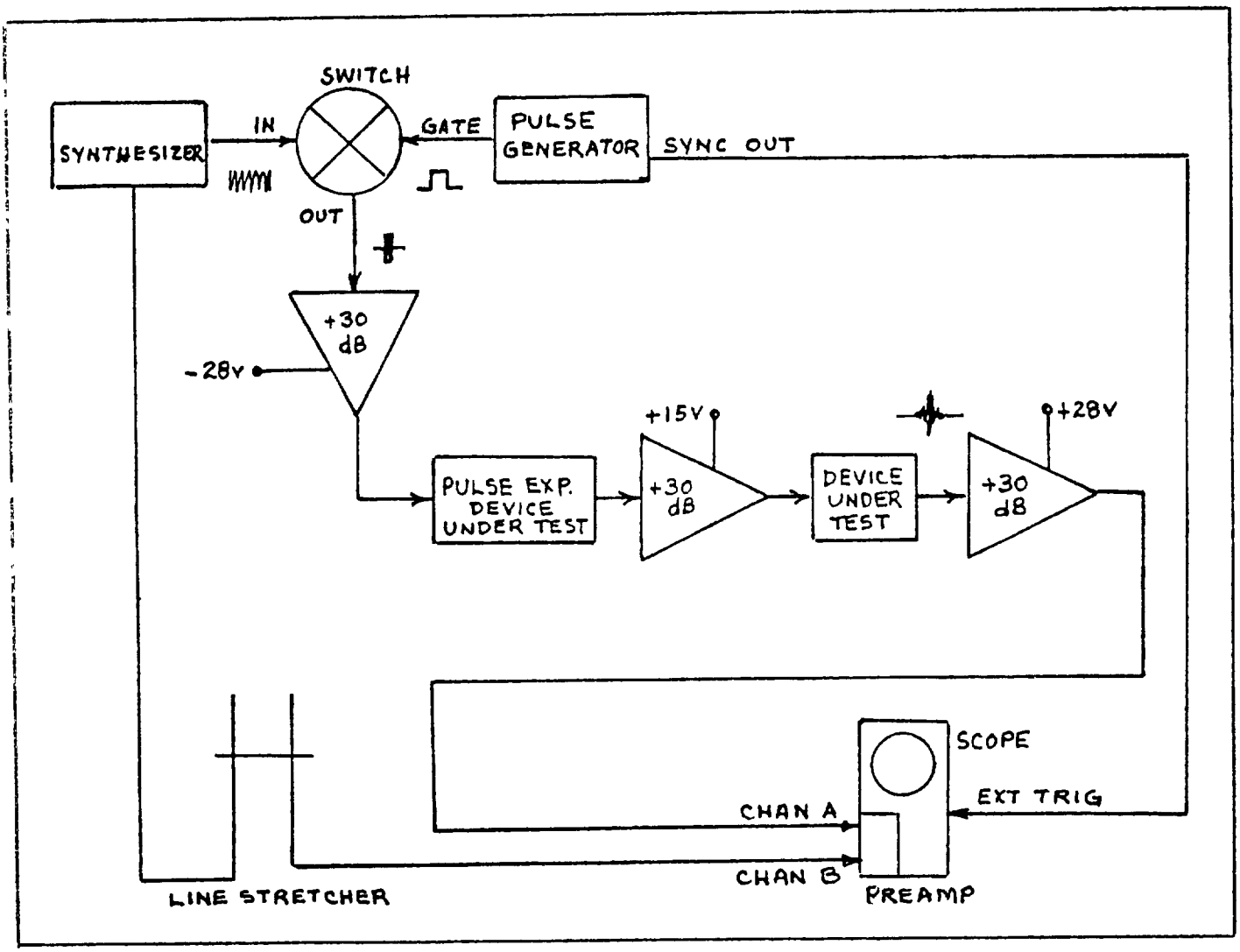


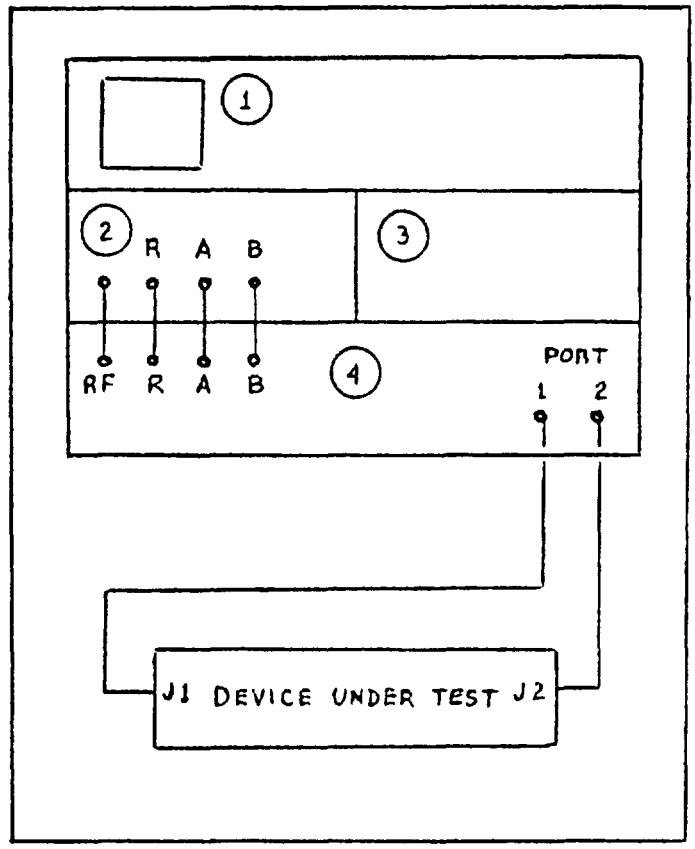
Figure 1. Block Diagram Measurement of Center Frequency (f_0) for a pulse compression (PC) matched SAW device pair. Amount of amplifier gain or possible inserted attenuation required in the channel containing the devices under test will vary with their insertion losses and the signal source output power levels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

Center frequency of bandpass filters is generally measured in the frequency domain. Using a network analyser in the transmission mode (Figure 2) with 1 dB/division amplitude sensitivity, the -3 dB points are measured on the high (F_{HI}) and low (F_{LO}) ends of bandpass. The center frequency is then calculated as:

$$F_C = \frac{F_{HI} - F_{LO}}{2} + F_{LO}$$

This measurement simultaneously measures the 3 dB bandwidth (β) of the filter.



Item	Description
1	Network Analyzer
2	Source Converter
3	Frequency Control Unit
4	S-Parameter Test Set

Figure 2. Block diagram for measurement of center frequency, bandwidth, insertion loss, and return loss in the frequency domain.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

6.4.2 Bandwidth (B -3 dB)

6.4.2.1 Bandwidth at the -3 dB points for frequency domain devices is normally measured as described above. The bandwidth of time domain devices is determined by the transmitted signal and is described below, for the case of pulse compression.

6.4.2.2 Bandwidth for pulse compression time domain devices. The generally accepted method of defining the operating parameters of a linear FM chirp device is to specify the instantaneous operating bandwidth and differential time delay of the transmitted signal. The time and frequency domain characteristics of a waveform having a bandwidth, β, and a differential delay, T, are shown in figure 3.

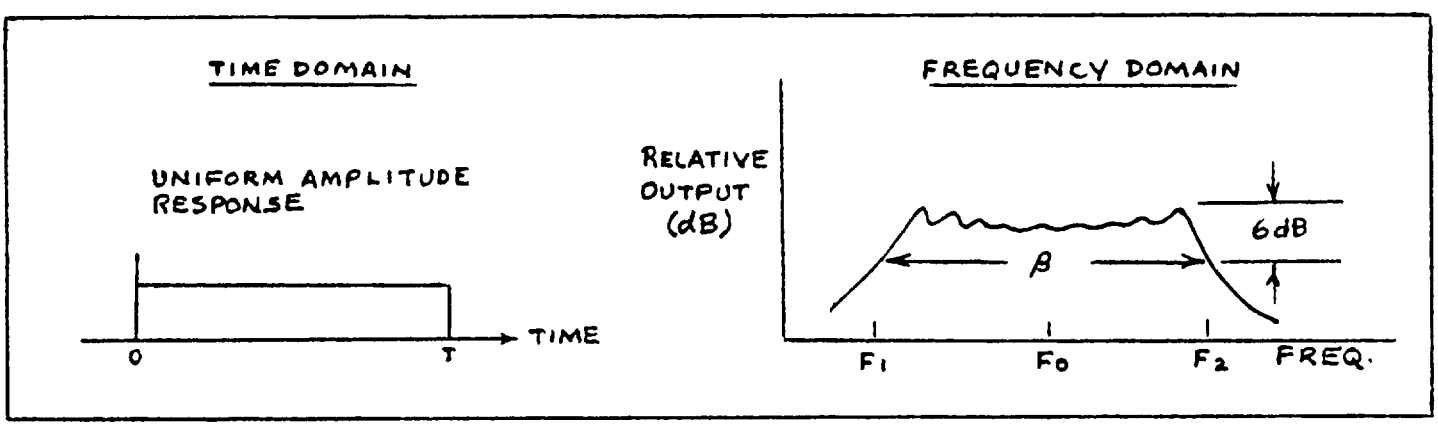


Figure 3. Time and frequency domain characteristics of a linear FM waveform

As shown in the figure, the time domain response is assumed to be flat over the pulse length, T. The instantaneous frequency of the carrier varies linearly between the extremes F_1 and F_2 which define the characteristic bandwidth to be $(F_2 - F_1)$. The spectrum of this waveform has a nominally rectangular envelope response whose -6 dB bandwidth is equal to (NOTE: If this waveform were correlated in a matched unweighted compression line having the opposite chirp slope, the recompressed pulse shown in Figure 4 would be obtained. With no weighting, the close in sidelobes would be 13 dB down from the peak output response. The -4 dB width of the recompressed pulse would be equal to 1/B while its null width would equal 2/B. The ratio of the transmitted pulse length to the -4 dB width of the recompressed pulse would be T/(1/B) or TB. Thus the pulse compression ratio of an unweighted chirp system is equal to the product of the transmitted pulse length times its instantaneous operating bandwidth.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

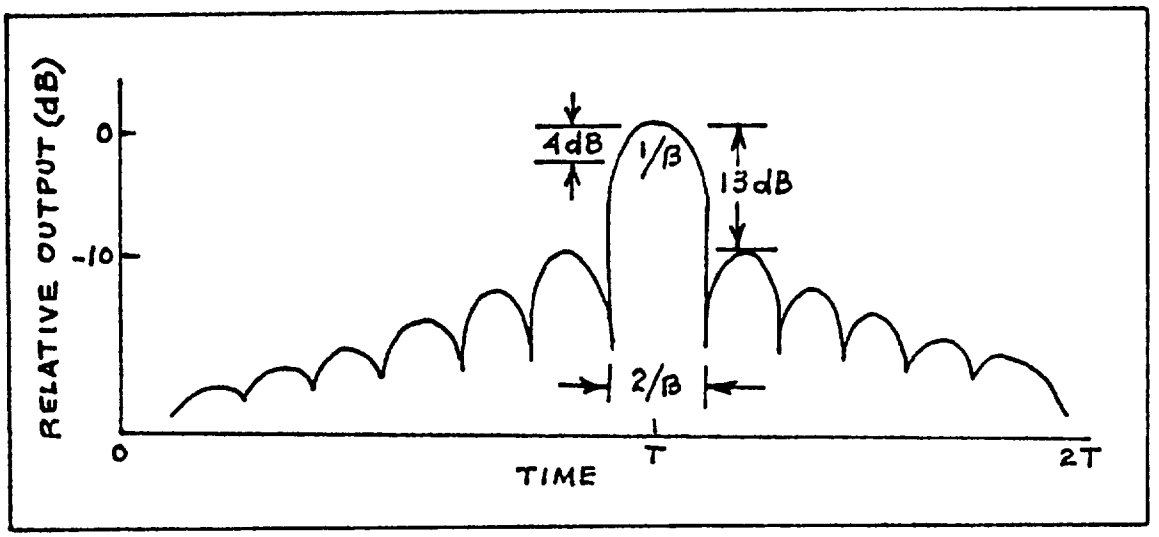


Figure 4. Recompressed Pulse Obtained with an Unweighted Compression Line

If the bandwidth is to be directly measured and not simply specified n the manner given above, it is generally measured in the frequency domain. For an unweighted device with a nominally flat passband amplitude response, the average insertion loss across the specified operating bandwidth is computed. Then the frequencies where the insertion loss drops and remains 3 dB below that average are measured. Their difference is the measured bandwith, and their arithmetic mean yields the center frequency for an individual PC device where the specified center frequency is not that of a matched pair of devices operating together. For a weighted device, where the passband amplitude response is not flat, the bandwidth is typically obtained by measuring the frequencies where the response drops and remains a prescribed amount below the peak, based upon the theoretical weighting function used, the nominal band edges are theoretically about 22 dB down from the center frequency peak. Thus the 22 dB frequencies would be measured to determine the bandwidth.

6.4.3 Insertion Loss-L. Insertion loss is most easily measured in the frequency domain for all devices types. It is measured by direct reading of the peak amplitude of the bandpass response of the device with respect to applied signal reference level. The standard transmission test setup shown in Figure 5 is used for this measurement. The applied signal reference is established by inserting a straight through connector in place of the device under test, and a horizontal line on the phase magnitude display unit is selected as the reference line. The vertical scale is set for a convenient scale factor such as 10 dB per division and the straight through connector is replaced by the device under test. The device response curve is then either measured at its peak or averaged over a defined bandwith. The difference between the resulting value and the reference is the device's insertion loss.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

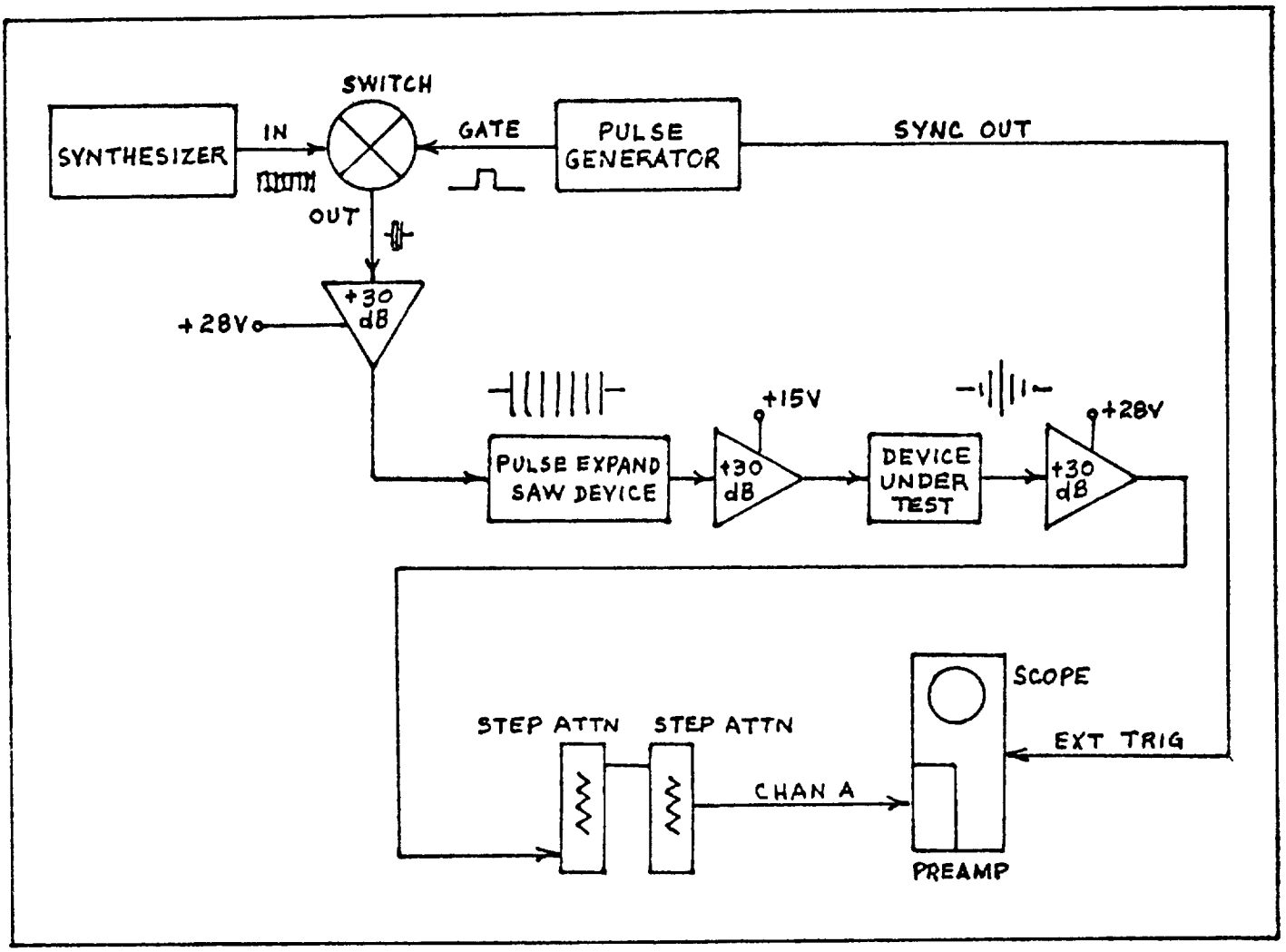


Figure 5. Block diagram for measurement of spurious and feedthrough suppression in the time domain, shown for the case of a compression line under test and with reference relative to the recompressed pulse.

6.4.4 Feedthough suppression. Feedthrough is a term used to express the amount of the original signal impressed upon the device under test which passes through the device without being altered by its transfer function. Since all signals processed by the device experience a time delay, the unprocessed feedthrough occurs earlier in time than the output response. An example can be seen in figure 6 for a time domain device. Feedthrough suppression is then the difference in amplitude between the desired output (the processed or delayed pulse) and the original unprocessed signal, expressed in dB. A typical value for feedthrough supression is 50 dB. As feedthough results from the input signal bypassing the device, it is associated with device packaging and time delay associated with transducer proximity. The measurement is made for devices by adjusting the attenuators to set the feedthrough response to a convenient amplitude on the oscilloscope

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

presentation and noting the magnitude of the feedthough pulse. Adjust the attenuators (figure 5) until the compressed pulse amplitude decreases exactly to the reference amplitude of the feedthrough pulse. The change in attenuator setting is the amount of feedthrough suppression.

For a pulse compression device, some care must be exercised when defining signal feedthrough suppression, because the measured level is specified relative to a different reference point in the case of a long input pulse or CW test setup than in a recompressed pulse test setup. In particular, when measuring relative to the main output response from a long input pulse (where the expansion loss is made negligible), the measurement value will not be the same as that observed when measured relative to the recompressed pulse mainlobe peak. The noted power level difference will be equal to the compression loop gain.

6.4.5 Spurious suppression-S-. Like feedthrough suppression, spurious suppression is most easily measured for all devices in the time domain. Spurious suppression is defined as the difference in amplitude between the desired signal and any undesired signal excluding feedthrough and sidelobes. In SAW devices, one major source of spurious signal is a result of reflections of the transmitted surface wave from the discontinuity represented by the transducer metalization. This type of spurious response has a precise relationship to the input, and is called triple transit spurious (TTS). Other spurious signals result from reflections occurring at the ends of the crystal, or other discontinuities. These sources of spurious response are generally minimized by placement of an absorbing material such as RTV on the top edges of the crystal, and/or angling the crystal ends. While these sources have a relationship to the desired signal, it varies with device design and crystal size and type. An example of TTS in the time domain can be seen in the oscilloscope trace in figure 6. Since sidelobes have a specific time relationship or duration with respect to the desired signal, spurious signals may be specified as undesirable signals so many microseconds beyond the carrier on either side. Sidelobes will be contained within the excluded time envelope. As an example, for a specific pulse compression filter those responses two microseconds and beyond on either side of the desired response are considered to be spurious. This is schematically depicted in figure 7. Spurious Suppression (or rejection) is measured by noting the difference between the largest spurious response and that of the desired signal. This is once again the change in step attenuator setting in figure 5 required to reduce the amplitude of the desired response to that of the largest spur. Refer to the paragraph on feedthrough suppression for detailed description of this measurement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

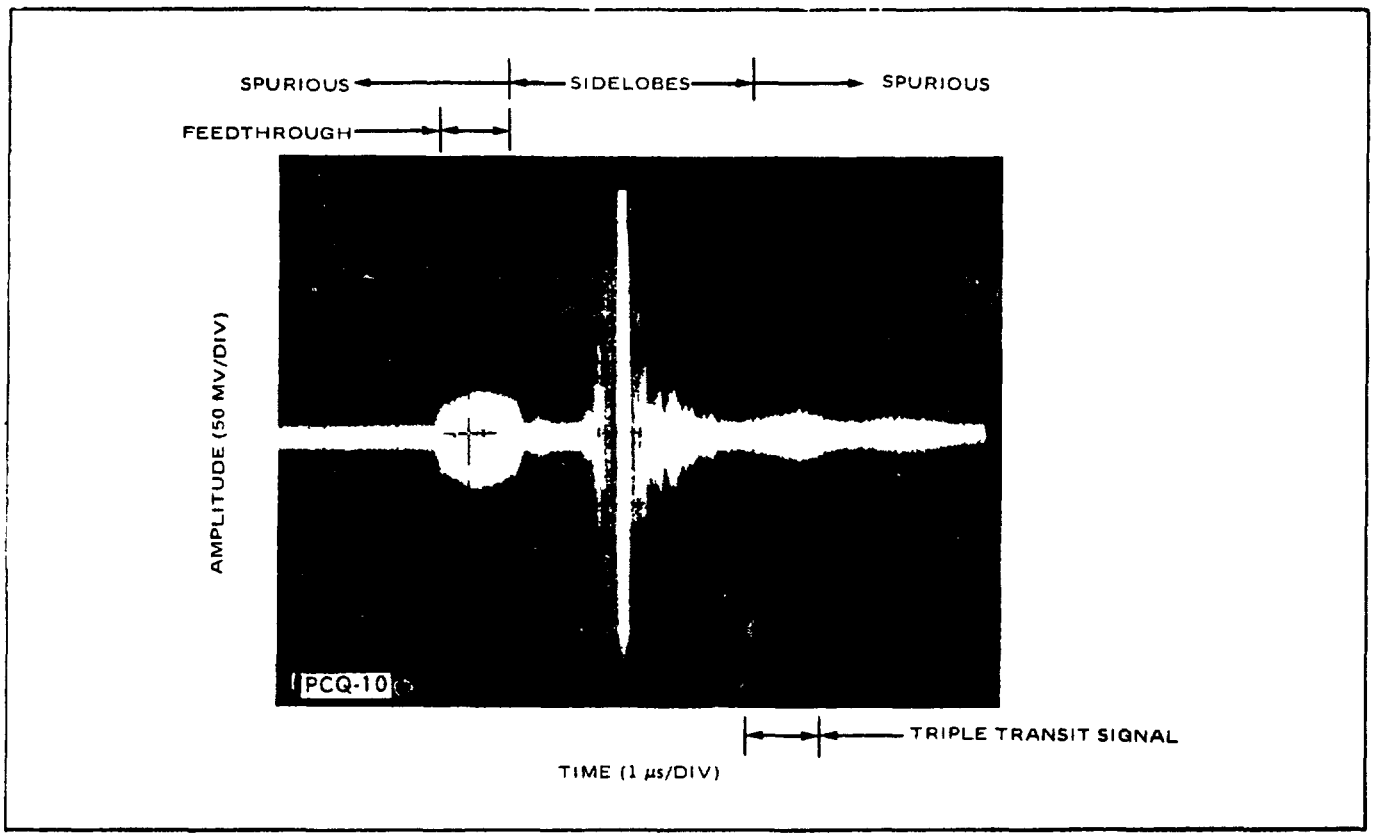


Figure 6. Typical spurious, feedthrough and sidelobe response for SAW delay line in the time domain

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

6.4.6 Voltage standing wave ratio - VSWR. This parameter is typically measured for all devices in the frequency domain. Two types of display are possible. The first of these is the Smith Chart Plot which depicts the real and imaginary impedance and the magnitude of the mismatch. The second display is the magnitude of the reflected energy (return loss) as a function of frequency. This magnitude of the return loss is readily converted into VSWR by use of standard charts and nomographs, and may be measured directly on a network analyzer. An example of the test setup is shown in figure 2.

Custodian:
Army - ER

Preparing Activity:
Army - ER

(Project 5999-A078)

Review Activity:
Army - AV

User Activity:
Army - MI

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems Enter in block 6 any remarks not related to a specific paragraph of the document If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
U.S. Army Electronics Research and
Development Command
ATTN: DELET-R-S
Fort Monmouth, NJ 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:00Z
Check the source to verify that this is the current version before use.